Corporate Finance – Business Valuation (Course 20127)

Free Cash Flow to Equity Models (FCFE)

Milan, October 28th-29th 2026

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Plan for the day

Last time

- Learning how to implement a DDM valuation
- Understanding the impact of Growth on a DDM valuation

Today

- Main questions for today:
  - Understanding FCFE Discount Models
  - Learning the Differences between DDM and FCFE

For Next Time

- Review the material covered in class

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Agenda

1. **Free-Cash Flows to Equity (FCFE)**
   a. From Earnings to Cash-Flows
   b. Comparing Dividends to FCFE
2. **FCFE Models**
   a. Principles
   b. Constant Growth Model
   c. Two-Stage Model
   d. Three-Stage Model
3. **Examples**
   a. Constant Growth Model: Sony
   b. Two-Stage FCFE Model: Nestlé

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From Earnings to Cash Flows

1. FCFE (Levered cashflows) – *Equity Side Valuation*

+ Net Income
+/- Non-Cash Expenditure / Non-Cash Revenues
- Capex
- Δ Non-Cash Non-Debt WC (WC in short now on)
- (Principal Debt Repayments - New Debt Issues)
= Free Cash Flows to Equity

2. FCFO (Unlevered cashflows) – *Asset Side Valuation*

+ FCFE
+ Interest Expense (1 – Tax Rate)
+ (Principal Debt Repayments - New Debt Issues)
= Free Cash Flows from Operations (FCFO)

+ Operating Income (EBIT)
- Taxes on EBIT
+/- Non-Cash Expenditure / Non-Cash Revenues
- Reinvestment Needs (CAPEX and WC)
= Free Cash Flows from Operations (FCFO)

[Ignoring preferred dividends. If preferred shares exist, preferred dividends will also need to be netted out]

FCFE: Dividends vs FCFE

- **Conventional measure of dividend policy (alternatively: *augmented* dividend payout):**

$$Payout = Dividends/Earnings$$

- **The *modified* approach measures <u>total</u> cash returned to shareholders as a % of FCFE (instead of net income):**

$$Cash\ to\ Shareholders = (Dividends + Buybacks) / FCFE$$

- **Three scenarios:**
  1. **<u>Ratio ≈ 1</u>:**
     - The firm is paying out <u>all</u> it can to shareholders
  2. **<u>Ratio < 1</u>:**
     - The firm is paying out <u>less</u> that it can and it is using the difference to increase cash or to invest in marketable securities
  3. **<u>Ratio > 1</u>:**
     - The firm is paying out <u>more</u> than it can and it is either drawing on existing cash or issuing new securities

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FCFE: Dividends vs FCFE

- **Many firms pay out <u>less</u> to shareholders than they have available in FCFE. Why? Plenty of reasons:**
  1. **<u>Desire for stability</u>:** firms are reluctant to change DPS which are "sticky" (variability in DPS < variability in EPS or cash flows)
  2. **<u>Signaling prerogatives</u>:** firms use DPS as signals of prospects, with increases viewed as positive signals and decreases as negative ones
  3. **<u>Future investments</u>:** firms might hold back on paying their entire FCFE as DIV if they expect substantial increases in CAPEX. Since issuing securities is expensive (transaction costs, ...), they may choose to keep excess cash to finance these future needs
  4. **<u>Managerial self-interest</u>:** managers may gain by retaining cash rather than paying it out as dividends

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FCFE: Principles

- **FCFE models discount potential dividends rather than actual ones. Thus, FCFE models are simple *variants* on the DDM**
- **Replacing dividends with FCFE assumes FCFE will be paid out to shareholders. There are two consequences:**
  1. **Cash:** No future cash build-up since the available cash after debt payments and reinvestments is assumed to be paid out to entirely
  2. **Growth:** Expected g in FCFE will include g in income from operating assets and not g in income from marketable securities
- **FCFE models we will analyze:**
  - ✓ Constant growth FCFE model
  - ✓ Two-stage FCFE model
  - ✓ Three-stage FCFE Model

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FCFE: Underlying Principles

- **FCFE are cashflows to equity investors. To estimate their growth:**

$$g = ERR \cdot ROE$$

- **For FCFE, it's better to replace RR with Equity Reinvest. Rate (ERR):**

$$ERR = \frac{(Capex - D\&A + \Delta\, WC - New\ Debt\ issued + Debt\ Repaid)}{Net\ income}$$

- **In the FCFE model, there is <u>no excess cash</u> and ROE should measure the return on noncash investment:**

$$Noncash\ ROE = \frac{Net\ Income - Aftertax\ interest\ income}{BV\ of\ Equity - Cash\ \&\ Marketable\ securities}$$

$$\boldsymbol{g = ERR \cdot Noncash\ ROE}$$

- **When discounting FCFE, it is safest to:**
  - ✓ Value the equity in operating assets
  - ✓ Add the existing cash the firm may have

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FCFE: Constant Growth model

- **Constant Growth FCFE is designed for firms in steady state:**

$$EqV_0 = \frac{\mathbb{E}(FCFE_1)}{K_e - g_n}$$

Where:

EqV_0	= Equity Value
$E(FCFE_1)$	= Expected FCFE next year
K_e	= Cost of equity (required return on equity)
g_n	= Expected growth rate of FCFE forever (stable g)

- **Stable growth rate:**
  - ✓ $g_n = ERR \cdot Noncash\ ROE$ (linked to fundamentals)
  - ✓ g has to be reasonable relative to the nominal g rate in the economy/industry the firm operates in
- **Limitations:**
  - ✓ Extremely (exponentially) sensitive to assumptions about g
  - ✓ If g is estimated from industry averages, the valuation can be skewed when the industry is at a different stage of growth (not in steady state) than the target firm

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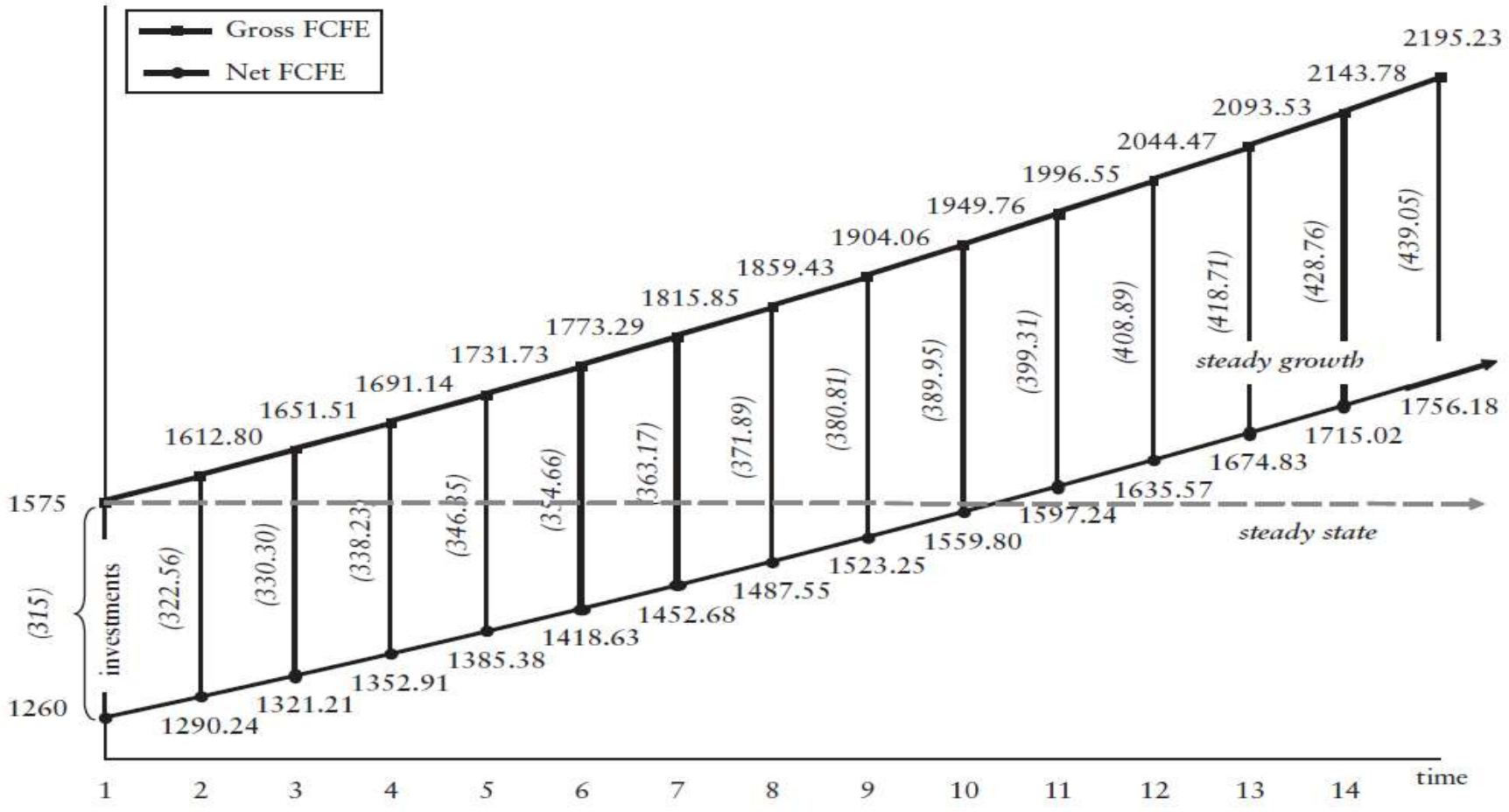
FCFE: Constant Growth model

- **Firms model works best for:**
  - ✓ Firm growing at a rate ≤ nominal growth in the economy
  - ✓ Better than DDM for firms paying dividends that are unsustainably high or are significantly lower than FCFE
  - ✓ If firms are stable and pay out their FCFE as dividends the value obtained will be the same as the one obtained from the Gordon Model

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FCFE: Constant Growth model

- **Cashflows of a hypothetical firm:**
  - ✓ Horizontal dashed: FCFE in a steady-state scenario
  - ✓ Assumption: In steady-state no new investments except renewals (CAPEX = D&A)



FCFE: Constant Growth model

- **What happens when the firm invests 20% of its yearly cashflows at a return of 12%?**
  - ✓ FCFE at T_1 is 1,260 (20% < FCFE in steady-state)
  - ✓ In T_2, Net Income is no longer 1,575, but 1,613 since the reinvestment of 315 in T_1 increases future Net Income by 38 (= 1,575 * 20% * 12%)
  - ✓ In T_2, again, the firm reinvests 20% of Net Income (323 = 1,613 * 20%)
  - ✓ In T_3, Net Income is no longer 1,575 but 1,652 (= Net Income in T_2 plus return on new investments) and so on to infinity
  - ✓ In a nutshell: cashflows distributable as dividends grow at a constant rate = 20% (ERR) * 12% (ROE) = 2.4%
- **Assuming Ke = 10%:**

$$EqV_0 = \frac{1{,}260}{0.10 - 0.024} = 16{,}579$$

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FCFE: Constant Growth model

- **In a *growth scenario*, equity value is equal to:**

  The base value calculated in a no-growth scenario (steady-state), plus

  The NPV of growth opportunities (NPVGO)

$$EqV_{Growth} = EqV_{Base} + NPVGO$$

- **The base value can be computed by discounting the steady-state scenario FCFE:**

$$EqV_{Base} = \frac{FCFE_{Steady-state}}{K_e} = \frac{1{,}575}{0.10} = 15{,}750$$

- **NPVGO = NPV of reinvestments:**
  - ✓ $NPVGO_1 = -315 + 38/0.10 = 63$
  - ✓ $NPVGO_2 = -323 + 39/0.10 = 65$
- **NPVGO can be thought of as a *growing perpetuity*:**

$$NPVGO = 63/(0.10 - 0.024) = 829$$

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FCFE: Constant Growth model

- **Equity value in the *growth scenario* is:**

$$EqV_{Growth} = \frac{1{,}575}{0.10} + \frac{63}{(0.10 - 0.024)} = 16{,}579$$

- **This is equal to using the *Gordon model* @ T_0:**

$$EqV_{Gordon} = \frac{FCFE}{(K_e - g)} = \frac{1{,}260}{(0.10 - 0.024)} = 16{,}579$$

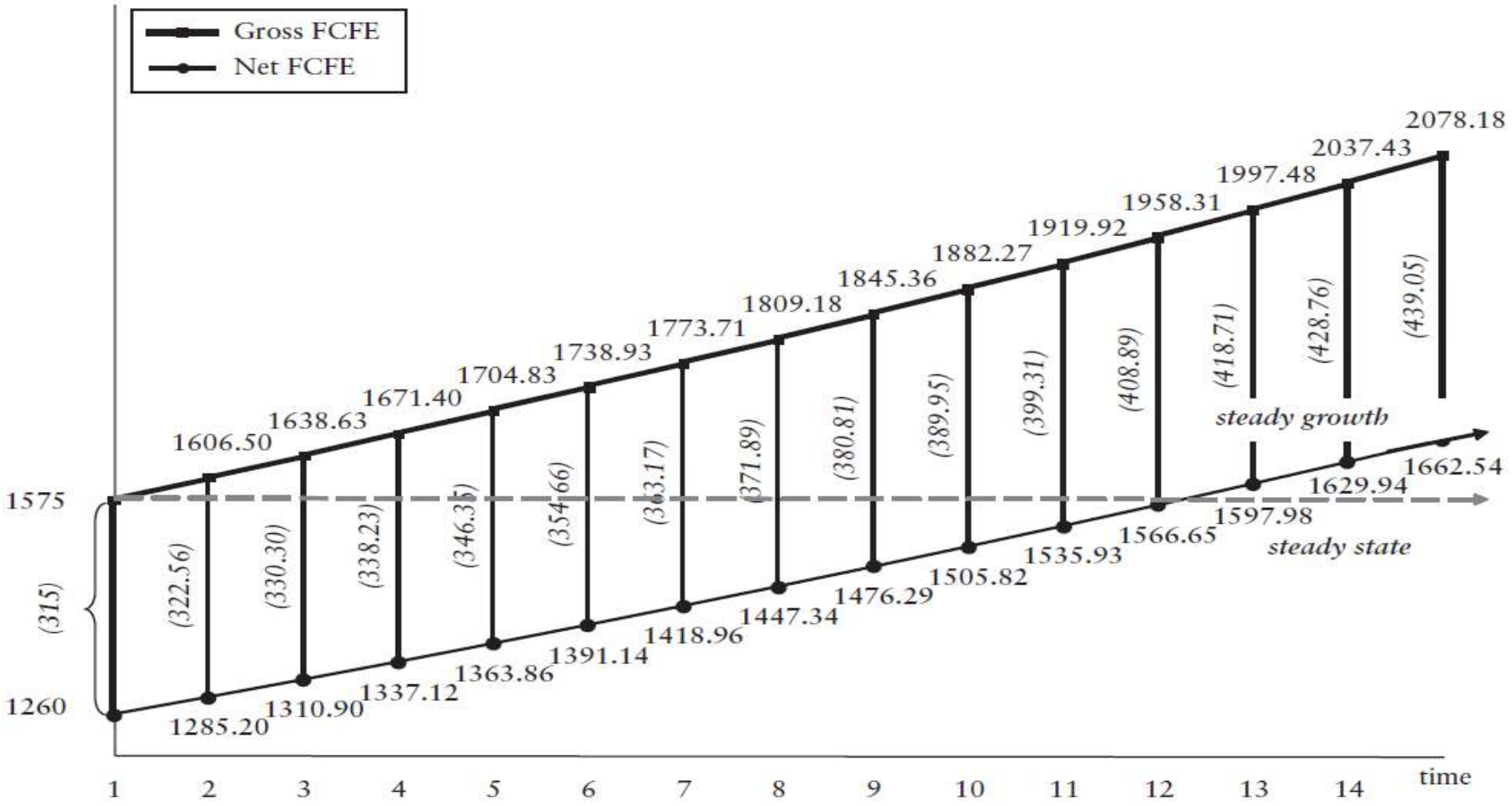
- **Growth only creates value if ROE > Ke (for asset side models, if ROI > WACC)**

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FCFE: Constant Growth model

- **Let's check what happens if (noncash) ROE = K_e = 10%**
- **To begin:**

$$g = ROE \cdot RR = 0.1 \cdot 0.2 = 0.02 = 2\%$$



FCFE: Constant Growth model

- **Let's determine equity value under the steady-state approach and the Gordon model**
  1. Steady-state approach:

$$EqV = \frac{1{,}575}{0.10} = 15{,}750$$

  2. "Gordon Model":

$$EqV = \frac{1{,}260}{(0.10 - 0.02)} = 15{,}750$$

- **Equity value is the same. Why?**
  - ✓ The firm grows in size (due to new investments and cashflows), but **not in value.** NPVGO are systematically null:

$$NPVGO\ T_1 = -315 + 31.5/0.10 = 0$$

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FCFE: Constant Growth model

- **If ROE < K_e → growth *destroys* value because NPVGO <0**

- **In our example, assume the new investment return is 8%**

$$g = 0.08 \cdot 0.2 = 1.6\%$$

- **As a result, the value computed with the Gordon model is:**

$$W = \frac{1{,}260}{(0.10 - 0.016)} = 15{,}000$$

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FCFE: Two-Stage Model

- **Two-Stage FCFE model is designed to value firms expected to grow faster than a stable firm at first and at a stable rate afterwards**

$$EqV = \sum_{t=1}^{t=n} \frac{\mathbb{E}(FCFE_t)}{(1+K_{e,hg})^t} + \frac{TV}{(1+K_{e,hg})^n}$$

$$= \sum_{t=1}^{n} \frac{\mathbb{E}(FCFE_t)}{(1+K_{e,hg})^t} + \frac{\mathbb{E}(FCFE_{n+1})}{(K_{e,st} - g_n)(1+K_{e,hg})^n}$$

Where:

$E(FCFE_t)$	= Free cash flow to equity at t
K_e	= Cost of equity in high (hg) and stable growth (st)
TV	= Terminal Value at the end of the "extraordinary" growth
g_n	= Stable Growth Rate

- **Stable Growth Rate:**
  - ✓ $g_n = ERR \cdot Noncash\ ROE$
  - ✓ g should be checked with the economy/industry growth rate

FCFE: Two-Stage Model

- **Limitations:**
  - ✓ Same limitations that apply to *g* for the stable growth model
  - ✓ Assumptions made to derive FCFE after the terminal year must be consistent with assumption of stability
  - ✓ Two approaches (same as in the stable growth model):
    1. Industry average CAPEX requirements
    2. Fundamental growth equation (ERR = g/ROE)
  - ✓ β and debt ratio may need to be adjusted since in stable growth $\beta \approx 1$ and debt ratio higher than during high-growth period

- **Firms model works best for:**
  - ✓ Firms that after a period of expansion are assumed to reach steady-state
  - ✓ Firms which either have unsustainable dividends (higher than FCFE) or paying less in dividends than they can afford to (less than FCFE)

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FCFE: Three-Stage Model

- **Three-Stage FCFE model is designed to value firms expected to go through three stages of growth:**

  1. Initial phase of higher g
  2. Transitional period in which g declines
  3. Steady-state period in which g is stable

$$EqV_0 = \sum_{t=1}^{t=n_1} \frac{\mathbb{E}(FCFE_t)}{(1+K_{e,hg})^t} + \sum_{t=n_1+1}^{t=n_2} \frac{\mathbb{E}(FCFE_t)}{(1+K_{e,tr})^t} + \frac{\mathbb{E}(FCFE_{n2+})}{(K_{e,st}-g_n)(1+K_e)^{n_2}}$$

Where:

$E(FCFE_t)$	= Expected free cash flow to equity in t
K_e	= Cost of equity (hg = high growth; tr = transition; st = stable growth)
n_1	= No. of periods in high growth stage
n_2	= No. of periods in transition stage
g_n	= Stable growth rate

- **Stable Growth Rate:**
  - ✓ $g_n = ERR \cdot Noncash\ ROE$
  - ✓ g should be checked with the economy/industry g rate

FCFE: Three-Stage Model

- **Limitations:**
  - ✓ Assumptions about all variables should be consistent with assumptions about the three phases of growth
  - ✓ Relationship between CAPEX and D&A as the firm goes from high *g* to stable *g* will change as follow:
    - *High-g phase:* CAPEX > D&A
    - *Transitional phase:* Difference between CAPEX and D&A narrows
    - *Stable-g phase:* Difference between CAPEX and D&A lower still
  - ✓ As *g* declines → $\beta \approx 1$

- **Firms model works best for:**
  - ✓ Firms with very high growth rates currently
  - ✓ Firms whose dividends are significantly higher or lower than the FCFE

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Constant Growth Example: Sony

- **Japanese firms have proved difficult to value. In fact:**
  - ✓ Earnings in LFY for most Japanese firms were depressed relative to earnings earlier in the decade, reflecting the Japanese economy
  - ✓ Japanese GAAPs understate earnings and overstate BV of equity, as firms are allowed to set aside provisions for unspecified expenses
  - ✓ Earnings of many export oriented Japanese firms tend be heavily influenced by FX movements
  - ✓ Cross holdings, and the lack of transparency in these holdings make it difficult to value them
- **Sony's key financials and market data are reported below:**

Figures in JPY	*20XX*
Revenue	2,593b
D&A	76b
Net income	38.23b
CAPEX	103b
WC	220b

Figures in JPY	*20XX*
BV equity	1,795b
Dividends	21b
ROE	2.13%
Risk-free	2.00%
ERP	4.00%

Constant Growth Example: Sony

- **To properly value Sony, we will:**
  - ✓ Normalize earnings using past Sony's ROE of 5.25% as current earnings are depressed
  - ✓ Dominant market share will allow Sony to benefit from high(er) growth → Constant Growth
  - ✓ Long-term stable g of 3% (> Japanese economy due to global exposure and equal to average g in revenues for Sony over the last 5 years)
  - ✓ CAPEX will grow at the same rate (3%) and WC will stay constant at 8.48% of revenues (= 220/2,593)
  - ✓ Reinvestment financed with a ratio equal to current debt to capital ratio of 25.8%
  - ✓ $\beta = 1.1$

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Constant Growth Example: Sony

- **Minority active holdings (between 20%-50%) raise valuation issues:**
  - ✓ Under the *equity method* of accounting, firms report, as part of Net Income but below EBIT, the share of net income they are entitled to from these holdings
  - ✓ This is irrespective of whether they received any cash distribution (dividends) from these minority holdings
  - ✓ If a firm has a minority passive holding in another firm (< 20%), then it only reports, as part of Net Income but below EBIT, in its P&L the cash dividends received
- **To value cross holdings correctly, one must:**
  1. Value the target firm without including cashflows from cross-holdings
  2. Value the cross-holdings on their own
  3. Take the % of the value of these companies owned by the target firm and add it on top of the target firm's valuation from Step 1

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Constant Growth Example: Sony

- **Normalized Earnings:**

$$Adjusted\ Net\ Income = 1{,}795 \cdot 5.25\% = 94.2$$

- **Reinvestment Needs:**
  - $Current\ Net\ CAPEX = 103 - 76 = 27$
  - $Expected\ Net\ CAPEX = 27 \cdot 1.03 = 27.8$
  - $Expected\ Revenues = 2{,}59 \cdot 1.03 = 2{,}67$
  - $Expected\ \Delta\ WC = (2{,}67 - 2{,}59) \cdot 8.48\% = 6.6$
- **Cost of equity:**

$$K_e = 2\% + 1.1 \cdot 4\% = 6.4\%$$

Expected FCFE next year (figures in JPY/b)	
Expected Net Income	94.2
- (Net CAPEX) · (1 – Debt Ratio) = 27.81 · (1 - .258)	(20.6)
- Δ WC · (1 – Debt ratio) = 6.6 · (1 - .258)	(4.9)
Free Cash Flow to Equity (FCFE)	**68.7**

$$EqV = \frac{\mathbb{E}(FCFE_1)}{k_e - g_n} = \frac{68.7}{.064 - .03} = 2{,}021$$

Two-Stage Growth Examples: Nestlé

- **Nestle has operations all over the world, with 97% of sales from markets outside Switzerland, where it is headquartered**
- **The firm, like many large European corporations, has:**
  - ✓ Weak corporate governance
  - ✓ Shareholders with little power over managers
  - ✓ Dividends < FCFE

- **Why Two-Stages?**

  Nestlé has a long and impressive history of growth and while we believe that its growth will be moderate, we assume that it will be able to maintain high growth for 10 years
- **Why FCFE?**

  Given its weak corporate governance structure and a history of accumulating cash, dividends paid by Nestlé bear little resemblance to what the firm could have paid out

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Two-Stage Growth Examples: Nestlé

Inputs	*CHF/M*	*Per share*
Revenues	81,422	2,095.8
D&A	3,330	85.7
Net income	5,763	148.3
CAPEX	5,058	130.2
WC	5,818	149.8
Δ WC	368	9.5
BV of equity	25,078	645.5
Net debt issues	272	7.0

- **Begin by estimating Ke in Swiss Francs:**
  - ✓ 10-year Swiss government CHF bond rate is 4%
  - ✓ $\beta = 0.85$
  - ✓ To estimate ERP, Nestlé's breakdown of revenues by region

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Two-Stage Growth Examples: Nestlé

- **Estimating ERP:**

CHF/B	*Revenues*	*Weight*	*ERP*
North America	20.2	24.8%	4.0%
South America	5.0	6.1%	12.0%
Switzerland	1.3	1. 6%	4.0%
Germany/France/UK	21.3	26.1%	4.0%
Italy/Spain	7.4	9.1%	5.5%
Asia	6.7	8.2%	9.0%
Rest of W. Europe	15.0	18.4%	4.0%
Eastern Europe	4.6	5.7%	8.0%
Total	**81.4**	**100.0%**	**5.3%**

- **The risk premium is 5.3%**
- **The cost of equity is:**

$$k_e = 4\% + 0.85 \cdot 5.26\% = 8.47\%$$

Two-Stage Growth Examples: Nestlé

- **Assume Ke will remain unchanged in perpetuity**
- **To estimate g, compute current FCFE:**

$$FCFE = Net\ income\ -(CAPEX - D\&A) - \Delta\, WC + Net\ debt\ issues$$
$$= 5{,}763 - (5{,}058 - 3{,}330) - 368 + 272 = CHF\ 3{,}939$$

- **ERR can be estimated as:**

$$ERR = 1 - \frac{FCFE}{Net\ Income} = 1 - \frac{3{,}939}{5{,}763} = 31.7\%$$

- **Noncash ROE and g are(1):**

$$ROE = 5{,}763/25{,}078 = 22{,}98\%$$
$$g = ERR \cdot ROE = .3165 \cdot .2298 = 7.3\%$$

- **In stable growth, assume:**
  - ✓ g = 4%
  - ✓ Ke unchanged
  - ✓ ROE dropping to 15%
- **ERR in stable growth is then:**

$$ERR_{sg} = \frac{g}{ROE} = \frac{4\%}{15\%} = 26.7\%$$

(1) = Assuming BV of equity does not include any cash holdings and net income does not include any interest income

Two-Stage Growth Examples: Nestlé

- **Compute PV of FCFE in high *g* period, assuming:**
  - ✓ Earnings, net CAPEX and WC growing at 7.27%
  - ✓ 33.92% of reinvestment needs coming from debt

  (*data shown below are per-share*)

Year	EPS	Net CAPEX	Δ WC(1)	Reinvest.	Debt Fin.	FCFE	PV
1	159.1	47.7	10.2	57.9	19.6	120.9	111.4
2	170.7	51.2	10.9	62.1	21.1	129.7	110.2
3	183.1	54.9	11.7	66.6	22.6	139.1	109.0
4	196.4	58.9	12.5	71.4	24.2	149.2	107.8
5	210.7	63.2	13.5	76.6	26.0	160.1	106.6
6	226.1	67.8	14.4	82.2	27.9	171.7	105.4
7	242.5	72.7	15.5	88.2	29.9	184.2	104.2
8	260.1	78.0	16.6	94.6	32.1	197.6	103.1
9	279.1	83.7	17.8	101.5	34.4	212.0	102.0
10	299.3	89.8	19.1	108.9	36.9	227.4	100.8
∑ PV of FCFE							**1,060.5**

(1) = Change in WC each year is computed based on the existing WC of CHF 149.76 per share.

Two-Stage Growth Examples: Nestlé

- **To estimate TV, estimate FCFE in year 11:**
  - ✓ $EPS_{11} = EPS_{10} \cdot (1 + g) = 299.3 \cdot (1.04) = 311.3$
  - ✓ $ERR_{11} = EPS_{11} \cdot Stable\ RR = 311.3 \cdot .2667 = 83.0$
  - ✓ $FCFE_{11} = EPS_{11} - ERR_{11} = 311.3 - 83.0 = 228.3$
  - ✓ $TV = FCFE_{11}/(Ke_{11} - g) = 228.3/(.0847 - .04) = 5{,}103.5$
- **Value per share can be estimated as:**
  - ✓ ∑ PV of FCFE in high growth phase
  - ✓ PV TV

$$EqV = 1{,}060.5 + \frac{5{,}103.5}{(1.0847)^{10}} = 3{,}323.3$$

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